

SHARWARI AKRE

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Software Engineer | Backend Systems • Agentic AI • LLM Applications

EDUCATION

M.S. in Computer Science

North Carolina State University, Raleigh, NC

08/2024 – 05/2026

GPA: 3.67 / 4.0

B.E. in Information Technology

Marathwada Mitra Mandal's College of Engineering, Pune, India

07/2019 – 05/2023

GPA: 3.8 / 4.0

TECHNICAL SKILLS

Languages: Python, Java, JavaScript/TypeScript, C++, SQL

ML & Data: scikit-learn, NumPy, Pandas, Librosa, PyTorch, CodeBERT, OpenAI API, Ollama, pgvector

Backend & Frontend: Node.js, Express, Django, FastAPI, Flask, React, Next.js, REST APIs

Databases & Cloud: PostgreSQL, MongoDB, AWS, GCP, Docker, Kubernetes, Linux

Tools & Practices: Git, Postman, Agile/Scrum, OpenAPI/Swagger, Jira, JMeter, Flawfinder, Twilio

WORK EXPERIENCE

Product Engineering Intern | Trust2Connect

05/2025 – 07/2025

- Eliminated API timeout issues under peak load and **reduced latency by 40%** by identifying and resolving connection-per-query bottlenecks via connection pooling.
- **Achieved sub-200 ms notification latency** by designing and deploying a Pub/Sub-based communication system.
- Enabled scalable, secure document storage across 10+ microservices using GCP Storage and AlloyDB (PostgreSQL).
- Deployed application services to a **Kubernetes**-based dev environment, enabling consistent team rollouts.
- Standardized API contracts using **Apigee**-integrated REST endpoints and OpenAPI/Swagger specs; engineered **React** components to consume and surface the APIs on the frontend.

Software Developer Trainee | Capgemini

12/2023 – 07/2024

- Applied indexing strategies and query optimization techniques to improve SQL and PL/SQL query performance.
- Automated **UNIX-based file operations** and logging workflows, reducing error rates and manual intervention.
- Collaborated on 5+ dashboards using **Power BI and Tableau**, enabling real-time KPI tracking across business units.

PROJECTS

Aria Companion - AI Voice Agent for Elderly Care | FastAPI, Twilio, Ollama (Llama3), Whisper, pgvector

[GitHub](#)

- Built a **proactive AI voice companion** that autonomously calls elderly individuals for daily check-ins, orchestrating an end-to-end pipeline with Twilio, STT (Faster-Whisper), TTS (Kokoro) and multi-turn LLM conversation management.
- Implemented **episodic memory** using LLM-based fact extraction and semantic retrieval (sentence-transformers + pgvector cosine similarity), enabling personalized conversations grounded in prior call history.
- Designed a real-time mood scoring system extracting acoustic features (energy, pitch, pause ratio, speech rate) via Librosa with a rolling per-user baseline and dual-channel SMS escalation triggered by LLM-detected distress or post-call mood regression.

CodeNarrator | FastAPI, Python, Ollama, D3.js

[GitHub](#)

- Developed an **autonomous codebase agent** that maps dependencies and self-directs exploration across 15 repos, cutting onboarding time from ~2 hours to under 5 minutes.
- Integrated a local **Qwen2.5-Coder 7B model via Ollama** to generate AI architectural summaries like component breakdowns and dependency explanations with no external API or cloud cost.
- Produced self-contained HTML reports with **interactive D3.js dependency graphs** for navigating unfamiliar codebases.

Agentic LLM Workflow for Automated Vulnerability Patch Generation | CodeLlama, CodeAstra, Python

[GitHub](#)

- Developed an **agentic dual-model LLM pipeline** to auto-generate patches for C/C++ vulnerabilities using semantic hints.
- **Reduced vulnerabilities by up to 50%** overall and 70% for high-severity CVEs across 92 Juliet Suite cases.
- Evaluated patches across security (Flawfinder), functionality (**>85% compilation success**), and quality (15–20% CodeBERT similarity gain over baseline) dimensions.

CERTIFICATIONS

AWS Certified Cloud Practitioner (CCP) — Amazon Web Services, 2026

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